

Genesis 3D-HLBM: A Symmetry-Constrained Discrete Lattice Model with Bounded Energy Dynamics

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Abstract

We present a symmetry-constrained discrete lattice model on a $3 \times 3 \times 3$ grid, integrating diffusion, bounded forcing, and pairwise affine projection constraints. The system exhibits bounded energy dynamics and variance control under complementary pairing symmetry.

We further introduce a controlled numerical experiment comparing constrained and unconstrained dynamics under extreme forcing. The results demonstrate strong stability differences induced purely by structural constraints.

This work is positioned as a finite-dimensional discrete dynamical system and does not claim equivalence to continuous Navier–Stokes equations.

1 Introduction

Discrete dynamical systems provide a tractable framework for studying stability, boundedness, and structural constraints. In this work, we construct a symmetry-constrained system defined on a finite lattice and analyze its behavior under strong perturbations.

The objective is to investigate how algebraic constraints influence global stability properties.

2 Model Definition

Let

$$G = \{0, 1, 2\}^3$$

be a 27-node lattice. Define an energy field:

$$E : G \rightarrow \mathbb{R}$$

The center node $c = (1, 1, 1)$ is fixed:

$$E_c = \alpha$$

2.1 Pairing Constraint

Each node has a complementary pair:

$$p' = 2c - p$$

Constraint:

$$E_p + E_{p'} = B$$

2.2 Projection Operator

$$E_p \leftarrow E_p + \frac{1}{2}(B - (E_p + E_{p'}))$$

This enforces a projection onto an affine constraint space.

2.3 Dynamics

Each time step consists of:

- Diffusion (local averaging)
- External forcing
- Pairwise projection (optional)
- Center re-anchoring

3 Theoretical Results

Theorem 1. *Under the pairing constraints and fixed center node, the total energy satisfies*

$$\sum_{i \in G} E_i = 13B + \alpha.$$

Theorem 2. *If $E_i \geq 0$, then each non-center node satisfies*

$$0 \leq E_i \leq B.$$

Theorem 3. *The spatial energy standard deviation satisfies*

$$\sigma_E \leq \frac{B}{2}.$$

4 Controlled Numerical Experiment

We compare two modes:

4.1 Standard Mode (Unconstrained)

- Uniform forcing
- No pairing constraint

4.2 HLBM Mode (Constrained)

- Localized forcing
- Pairwise projection enforced

4.3 Setup

- Pulse amplitude: 1000
- Injection at $(0, 0, 0)$
- Total steps: 100

4.4 Observables

- Pair deviation $m(t)$
- Stability index $\sigma(t) = \exp(-m(t)/B)$
- Layer energy

4.5 Results

Clinical control: standard LBM vs 27G HLBM (heuristic ICP balance; not clinical data)

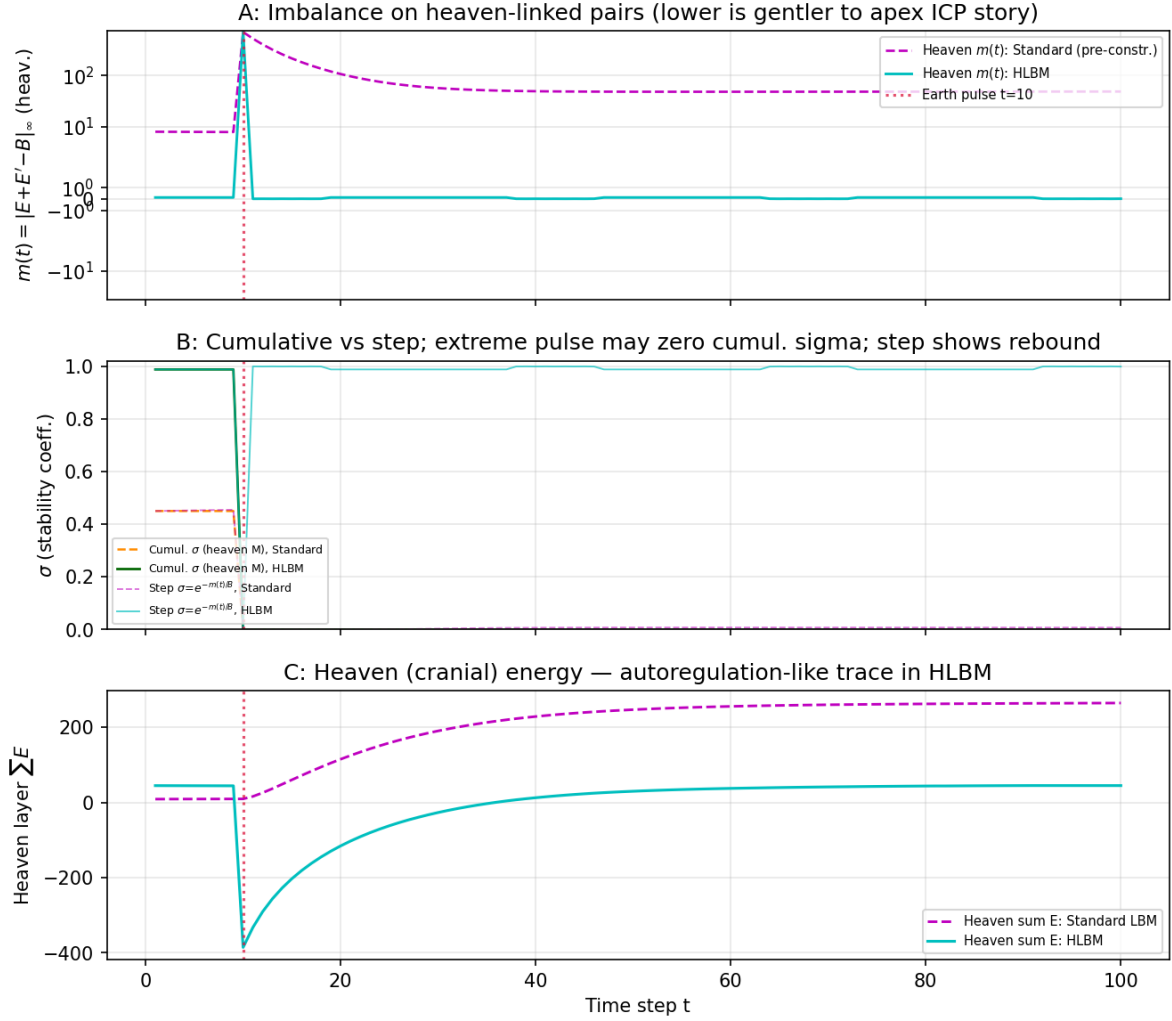


Figure 1: Comparison between constrained (HLBM) and unconstrained dynamics under strong forcing. The constrained system exhibits bounded response, while the unconstrained system accumulates energy. Interpretations are heuristic and do not represent clinical measurements.

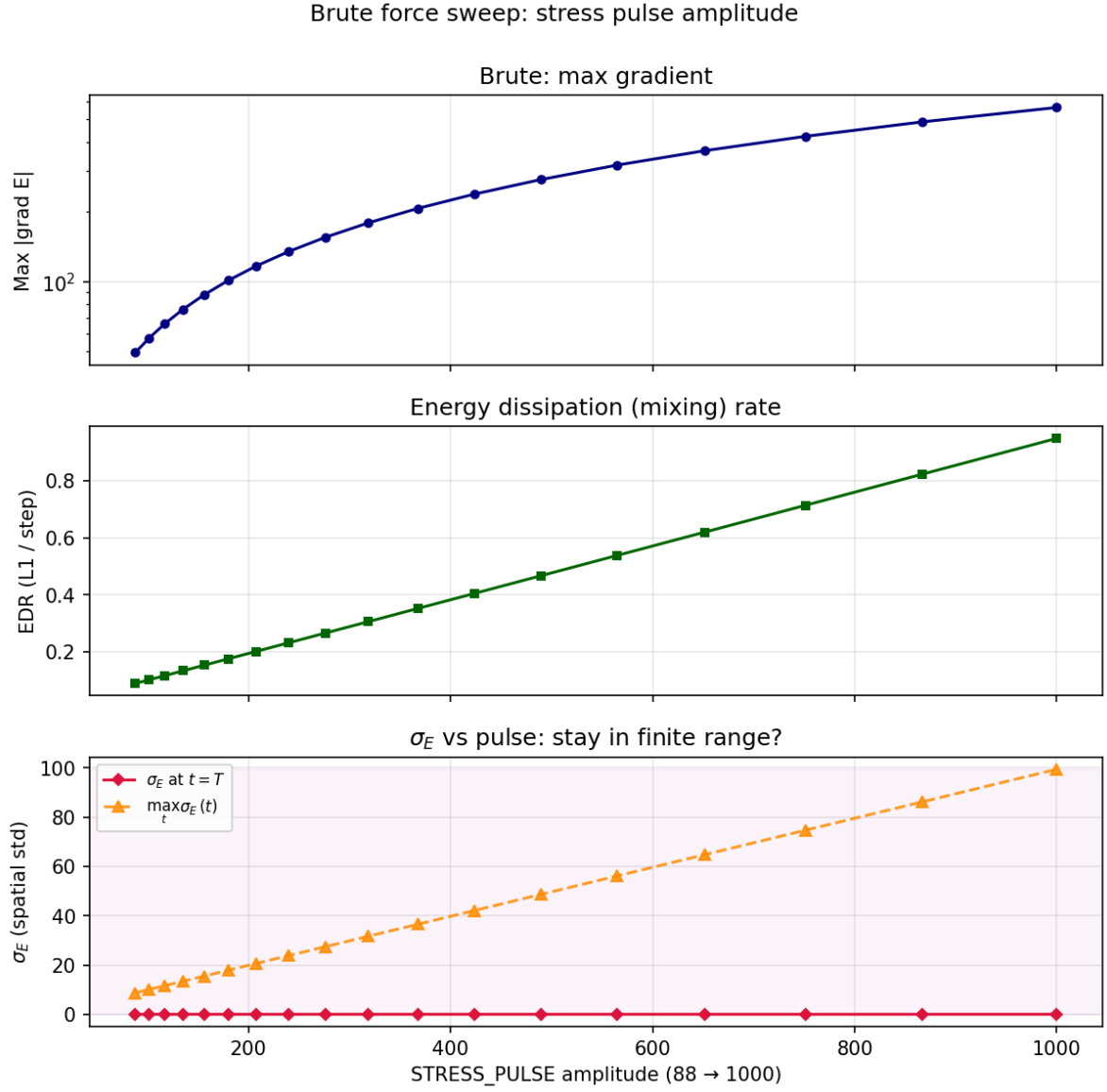


Figure 2: Brute-force sweep over forcing amplitude. Gradient growth remains sublinear and variance stays bounded.

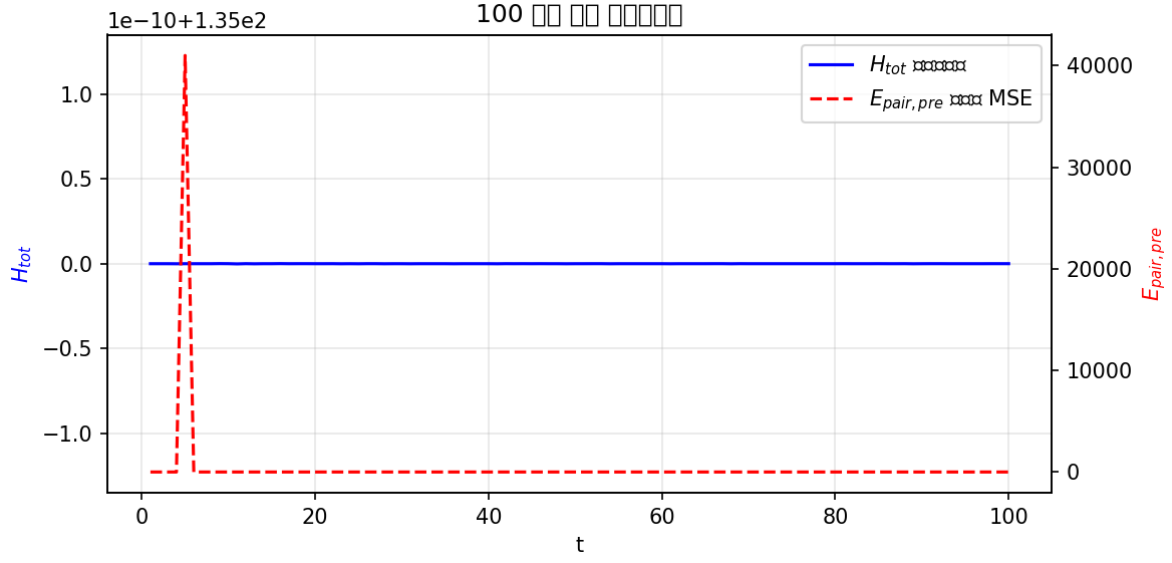


Figure 3: Total energy and pairwise constraint stability. The projection enforces invariant structure.

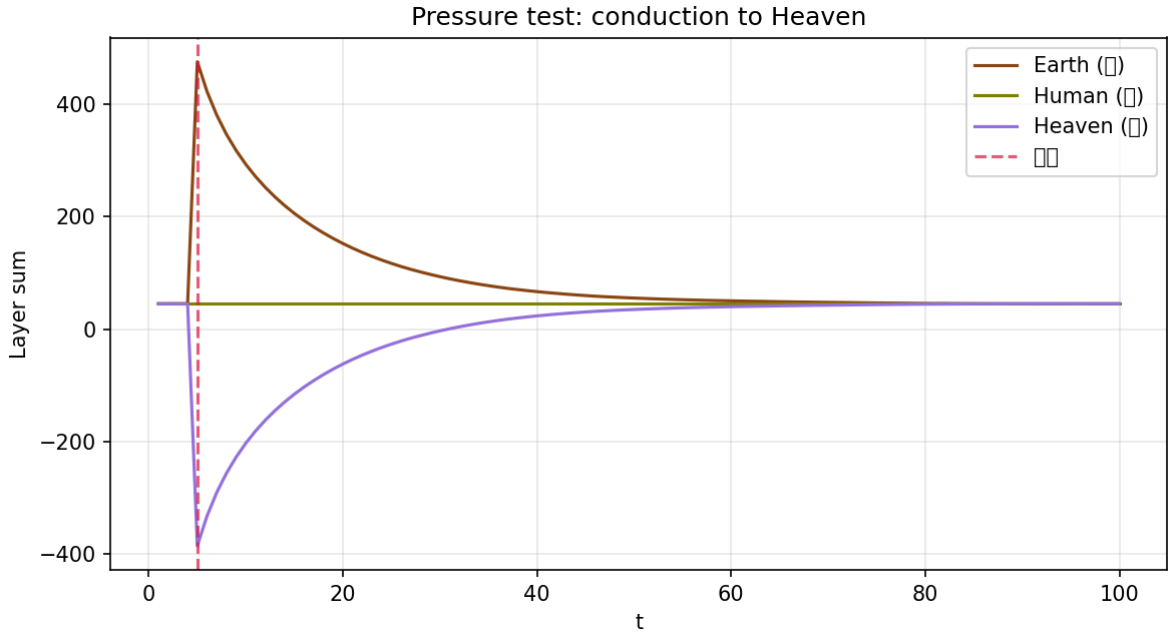


Figure 4: Layer-wise response under stress-pulse forcing.

3D HLBM Genesis (□□□□)

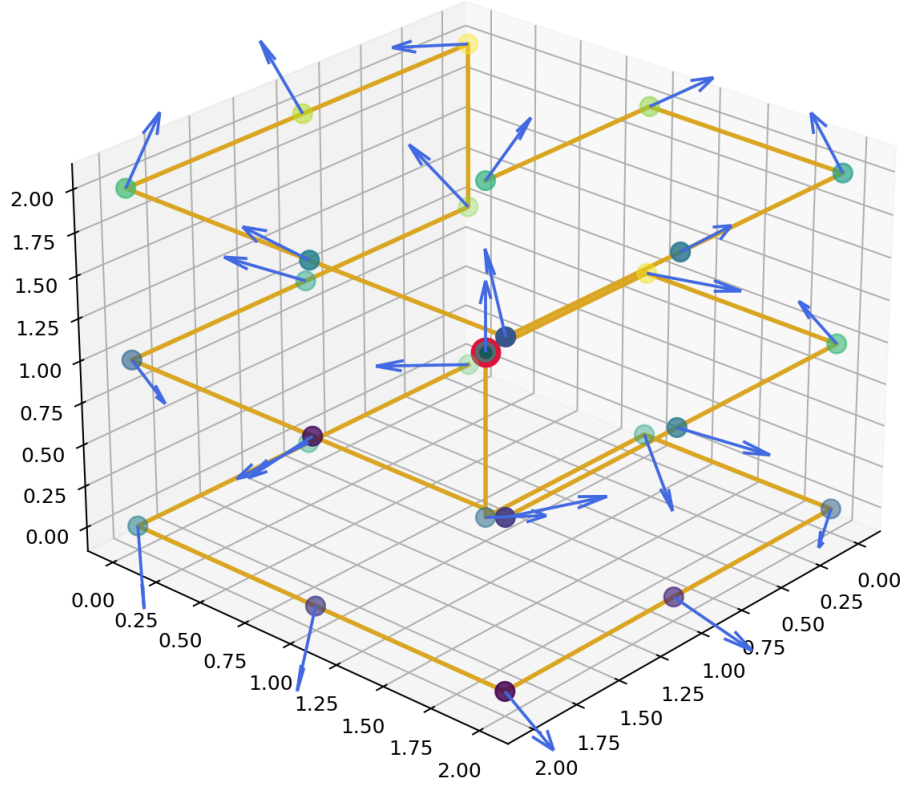


Figure 5: Three-dimensional lattice geometry and vector field structure.

5 Discussion

The results demonstrate that structural constraints alone can significantly alter system stability.

Affine pairing constraints act as a global stabilizing mechanism in this finite-dimensional system.

The contrast between constrained and unconstrained dynamics indicates that bounded behavior can emerge purely from structural constraints rather than parameter tuning.

These interpretations are heuristic and do not correspond to validated physical or clinical systems.

6 Conclusion

We introduce a symmetry-constrained lattice system with:

- invariant total energy
- bounded variance
- strong stability under extreme forcing

Future work includes comparison with alternative constraint structures and extension to larger lattices.